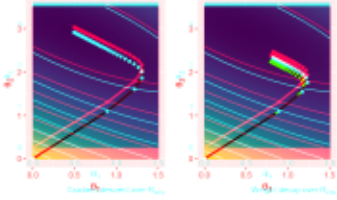
Introduction to Machine Learning

Regularization
Weight Decay and L2



Learning goals

- *L2* regularization with GD is equivalent to weight decay
- Understand how weight decay changes the optimization trajectory

WEIGHT DECAY VS. L2 REGULARIZATION

Let's optimize *L2*-regularized risk of a model $f(\mathbf{x} \mid \boldsymbol{\theta})$

$$\min_{\boldsymbol{\theta}} \mathcal{R}_{\text{reg}}(\boldsymbol{\theta}) = \min_{\boldsymbol{\theta}} \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) + \frac{\lambda}{2}\|\boldsymbol{\theta}\|_2^2$$

by GD. The gradient is

$$\nabla_{\boldsymbol{\theta}} \mathcal{R}_{\text{reg}}(\boldsymbol{\theta}) = \nabla_{\boldsymbol{\theta}} \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) + \lambda \boldsymbol{\theta}$$

We iteratively update $\boldsymbol{\theta}$ by step size α times the negative gradient

$$\begin{aligned}\boldsymbol{\theta}^{[\text{new}]} &= \boldsymbol{\theta}^{[\text{old}]} - \alpha \left(\nabla_{\boldsymbol{\theta}} \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}^{[\text{old}]}) + \lambda \boldsymbol{\theta}^{[\text{old}]} \right) \\ &= \boldsymbol{\theta}^{[\text{old}]}(1 - \alpha\lambda) - \alpha \nabla_{\boldsymbol{\theta}} \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}^{[\text{old}]})\end{aligned}$$

We see how $\boldsymbol{\theta}^{[\text{old}]}$ decays in magnitude – for small α and λ – before we do the gradient step. Performing the decay directly, under this name, is a very well-known technique in DL - and simply *L2* regularization in disguise (for GD).

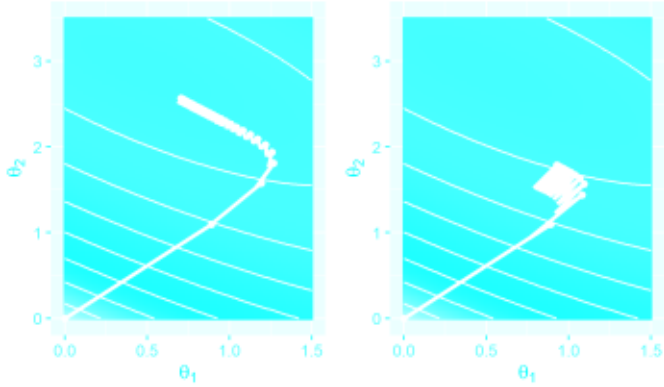
CAVEAT AND OTHER OPTIMIZERS / 2

Caveat: Equivalence of weight decay and *L2* only holds for (S)GD!

- Loshchilov and Hutter 2019 originally define WD "decoupled" from gradient-updates $\alpha \nabla_{\boldsymbol{\theta}} \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}^{[\text{old}]})$ as
  $\boldsymbol{\theta}^{[\text{new}]} = \boldsymbol{\theta}^{[\text{old}]}(1 - \lambda') - \alpha \nabla_{\boldsymbol{\theta}} \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}^{[\text{old}]})$
- This is equivalent to modern WD/*L2* (last slide) using reparameterization $\lambda' = \alpha \lambda$
- Consequence: if there is optimal λ', then optimal *L2* penalty is tightly coupled to α as $\lambda = \lambda' / \alpha$ (and vice versa)
- Loshchilov and Hutter 2019 show no equivalence of *L2* and WD possible for adaptive methods like Adam (Prop. 2)
- In many cases where SGD+*L2* works well, Adam+*L2* underperforms due to non-equivalence with WD
- They propose a variant of Adam decoupling WD from gradient updates (AdamW), increasing performance over Adam+*L2*

WEIGHT DECAY VS. L2 REGULARIZATION / 3

How strongly we are pulled back (for fixed α) depends on λ:



Small λ Large λ

CAVEAT AND OTHER OPTIMIZERS

Caveat: Equivalence of weight decay and $L2$ only holds for (S)GD!

- ▸ Hanson and Pratt 1988 originally define WD "decoupled" from gradient-updates $\alpha \nabla_\theta \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}^{[\text{old}]})$ as $\boldsymbol{\theta}^{[\text{new}]} = \boldsymbol{\theta}^{[\text{old}]}(1 - \lambda') - \alpha \nabla_\theta \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}^{[\text{old}]})$
- This is equivalent to modern WD/$L2$ (last slide) using reparameterization $\lambda' = \alpha\lambda$
- Consequence: if there is optimal λ', then optimal $L2$ penalty is tightly coupled to α as $\lambda = \lambda'/\alpha$ (and vice versa)
- ▸ Loshchilov and Hutter 2019 show no equivalence of $L2$ and WD possible for adaptive methods like Adam (Prop. 2)
- In many cases where SGD+$L2$ works well, Adam+$L2$ underperforms due to non-equivalence with WD
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